Learning Orbital Rendezvous

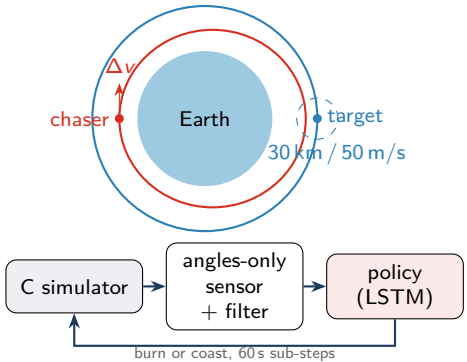
A reinforcement-learned guidance policy for a chaser spacecraft

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One policy, seven rendezvous skills in 3-D with J2, 96–99.5 % held-out success on the five wide-box skills.
Simulator in C, policy in PyTorch, every number measured on held-out episodes.

The problem, and the loop that solves it



Goal get a chaser inside 30 km and 50 m/s of a passive target

Ends when success, or re-entry, escape, empty tank, or the clock runs out

Budget 478 m/s of Δv (15 % propellant, Isp 300 s); about 2.2× a Hohmann transfer across the band, so no brute force

Start random LEO orbits, any phase gap up to $\pm 180^\circ$, independent eccentricity and apse orientation

Sees 38 floats: own elements, fuel, the estimated target, a target-relative block, the clock

Does one of 31 discrete actions: an impulsive burn, or a coast from 1 min to 24 h

Target state comes from line-of-sight angles only (batch acquisition, then a recursive filter); the policy flies on that estimate.

The physics: analytic two-body, and why a lower orbit catches up

State $(a, e, i, \Omega, \omega, M)$; Cartesian only when needed

Propagate $M \mathrel{+}= n\,\Delta t$ with $n = \sqrt{\mu/a^3}$, then Kepler's equation by Newton

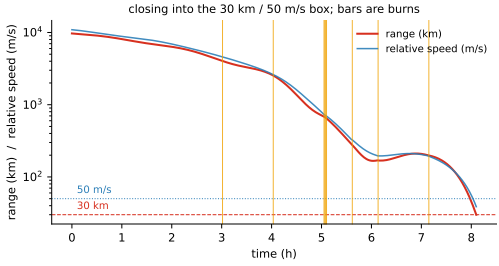
$$E \leftarrow E - \frac{E - e\sin E - M}{1 - e\cos E}, \qquad \tan\frac{\theta}{2} = \sqrt{\frac{1+e}{1-e}}\tan\frac{E}{2}$$

Elements $\leftrightarrow$ **Cartesian** perifocal frame, then $R_3(\Omega)R_1(i)R_3(\omega)$; inverse from $\mathbf{h} = \mathbf{r} \times \mathbf{v}$ and $\mathbf{e}$ directly

Burns impulsive along $\hat{v}, \hat{r}, \hat{h}$; mass by the rocket equation

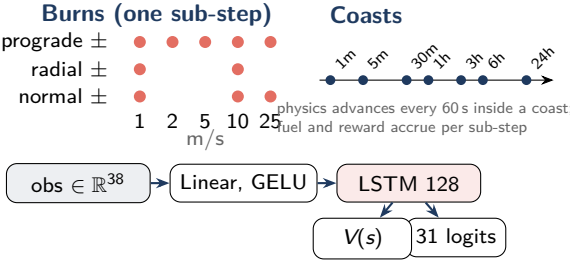
J2 secular drift of Ω, ω, and the clock

Why analytic exact at any Δt, so a 24 h coast is one solve, not 1,440 integrator steps. Roughly half a million env steps per second on one laptop CPU.



Phasing $T = 2\pi\sqrt{a^3/\mu}$: 100 km lower gains about 4° per orbit. Drop, wait, raise. The policy rediscovered this; the burns above are it doing so.

Decisions: a small recurrent policy over burns and coasts, trained with PPO



Semi-Markov one decision spans 60 s to 24 h; the headline policy takes **24 decisions** per episode. Without coasts the credit horizon is thousands of steps and nothing learns.

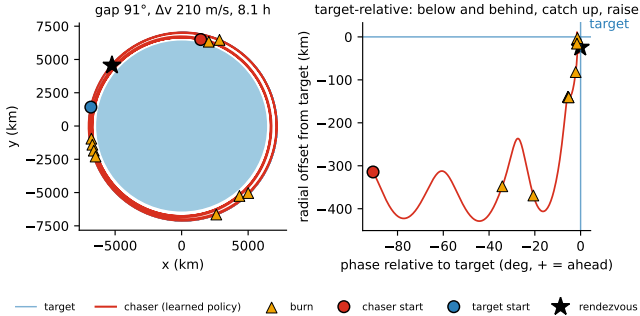
Why an LSTM what a burn did, and whether the estimate is settling, live in the history, not the current frame. 139,281 parameters.

PPO 1024 parallel envs, $\gamma = 0.995$, clipped surrogate ± 0.2, GAE, entropy bonus; 50 M steps per rung, about 2 h each on a laptop.

Reward $+1$ arrive, -1 fail, plus a potential-difference hint per decision on the mean-longitude gap and element match, clamped to $[-1, 1]$.

Curriculum warm-start one variable per rung: circular at the full gap $\rightarrow$ eccentric $\rightarrow$ wide envelope $\rightarrow$ 3-D $\rightarrow$ estimate in the loop $\rightarrow$ seven-skill mixture.

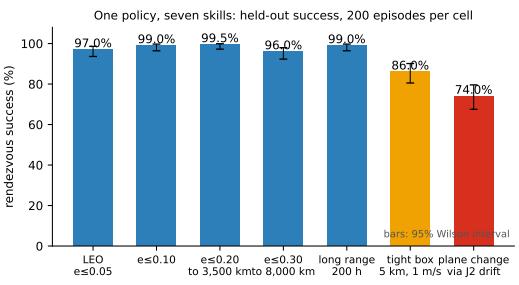
What it learned, 2-D: every held-out episode rendezvous, on every training seed

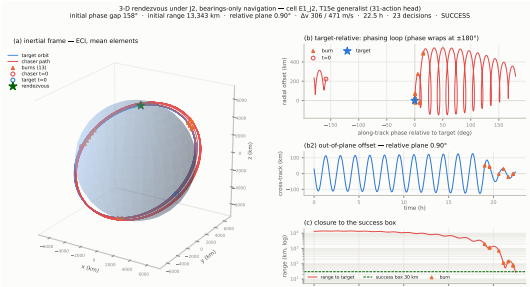


- **200/200** held-out episodes for **5/5 training seeds**, including the seed that flat-lined at 0 % before the dynamics fix.
- Re-run for this deck on a fresh evaluation seed: 200/200.
- Wide envelope (300–8,000 km, $e \leq 0.30$): 4 seeds, 800/800.

A 91° gap closed in 8.1 h with 210 m/s: below and behind, let the lower orbit catch up, then raise in steps. Median fuel is 1.18× a two-impulse optimum on the same clock (appendix).

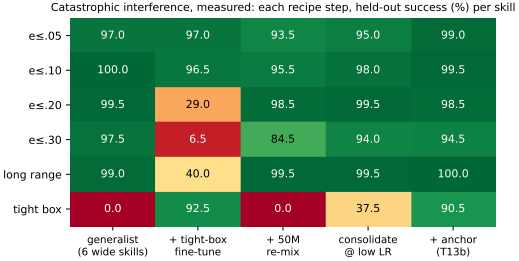
One policy, seven rendezvous skills, 3-D with J2

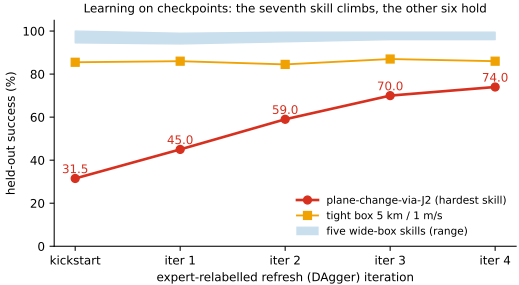




Left: held-out success per skill, 200 episodes each, bearings-only navigation; six of the seven skills reproduce at a second training seed; tight box reaches 92.5 % at its training fuel floor. Right: one held-out episode, a 158° gap and a 0.9° plane difference closed in 22.5 h with 306 m/s and 23 decisions. The seventh skill waits for J2 nodal precession to close a plane gap that burning could not afford.

How it got there: interference measured, then fixed





- Fine-tuning for the tight box destroyed three other skills; re-mixing restored them and lost the tight box. Low-learning-rate consolidation with an **anchor** (replay of the specialist's own trajectories) held all six.
- The seventh skill never bootstrapped from reward. It came from four rounds of **expert-relabelled refresh** against a scripted expert, with a stopping rule fixed in advance; reproduced at a second seed.

What I caught, what is open, what is next

Caught

- A swapped half-angle factor in the true-to-mean anomaly conversion let every burn teleport phase (median 25° per episode). The old headline fell from 97.5 % to 13 % once fixed; everything here is post-fix.
- A reward leak that made coast-only policies look capable at high altitude (null test: a zero policy reproduced it).
- A numbered ledger of metric-versus-implementation bugs, each with a before/after number.

Also built a two-impulse Lambert comparator for the fuel audit; a bearings-only navigation stack; an allocation-free C port of the policy, action-for-action identical to PyTorch.

Open

- The seventh skill's final refinement chain is one training seed; six skills reproduce at a second seed.
- Fuel is 1.18× the two-impulse optimum, not 1.0; the burn grid is coarse.
- Chaser navigation assumed perfect; no eclipses or sensor bias; arrival certified, not station-keeping.

Next

- A matched MPC comparator; a maneuvering target.
- Finer burn grades; multi-seed the generalist.

Appendix

observation vector · J2 and phasing · the seven cells · the dynamics bug · fuel audit · navigation in one picture · C port · if asked

A1 Observation vector (38 floats, roughly in $[-1, 1]$)

slots	value	why
0–1	chaser altitude / scale, e	size and shape
2–3	$\sin\theta, \cos\theta$	position on orbit, no wrap
4–5	v_r/v_{circ}, v_t/v_{circ}	radial and tangential speed
6	fuel fraction	budget remaining
7–8	target altitude, e (estimated)	target size and shape
9–12	$\sin, \cos\ \omega$ for both	apse orientations
13–14	$\sin, \cos\ \Delta\lambda$ (mean-longitude gap)	the phase gap
15	(cap − step)/cap	episode clock
16	$\cos(\omega_s - \omega_t)$	apsidal alignment
17–20	nearest body: distance, bearing, closing rate, keep-out	Earth
21–25	relative-inclination vector; $\Delta\mathbf{e} \cdot \hat{h}_t$; cross-track offset and rate	3-D block
26–28	Δv_{plane}, $\Delta v_{remaining}$, feasibility margin	fuel ledger
29–30	$\cos i_s$, $\cos i_t$	J2
33–36	LVLH radial / along-track offset; relative velocity, rotating-frame corrected	relative state
37	target mean motion	

Target quantities are the filter's estimate in bearings-only mode. The policy is not given the covariance (tested: a σ channel added nothing).

A2 Plane change by burning versus by J2

By burning $\Delta v = 2v\sin(\Delta i/2)$. At 7.5 km/s a 3° change costs about 390 m/s, most of a 478 m/s tank.

By J2 Earth's equatorial bulge makes the orbit's node line drift around the pole at

$$\dot{\Omega} = -\tfrac{3}{2}\, J_2\, n \left(\frac{R_E}{p}\right)^2 \cos i, \qquad p = a(1-e^2).$$

The rate depends on altitude and inclination, so two orbits at different altitudes drift apart in Ω at a controllable relative rate. Dip low, coast a day at a time, let the node drift close the plane gap, then raise. In the specialist about half of the plane alignment was bought this way; the rest with normal burns near the relative node.

Phasing $T = 2\pi\sqrt{a^3/\mu}$; a chaser 100 km below a LEO target gains about 4° of phase per orbit, so "drop, wait N orbits, raise" is the textbook catch-up and it is what the policy does.

A3 The seven cells

cell	altitude band (km)	e_{max}	box	clock (sub-steps)	success
E0_j2	300–800	0.05	30 km / 50 m/s	3,000	97.0 %
E1_j2	300–1,500	0.10	30 km / 50 m/s	3,000	99.0 %
E2_j2	300–3,500	0.20	30 km / 50 m/s	4,500	99.5 %
E3_j2	300–8,000	0.30	30 km / 50 m/s	6,000	96.0 %
LONGRANGE	300–3,500	0.12	30 km / 50 m/s	12,000	99.0 %
TIGHT_5k1	300–800	0.05	5 km / 1 m/s	3,000	86.0 %†
W1_driftwait	300–800	0.05	30 km / 50 m/s	22,000	74.0 %

†92.5 % at the training fuel floor. All cells: 3-D, J2 on, targets inclined 30–60°, relative plane up to 0.75–1° (2–5° for W1), bearings-only navigation, Discrete(31) head, fuel budget sampled per episode (353–656 m/s). Held-out seed 123, 200 episodes per cell, greedy actions, LSTM state zeroed per episode.

A4 The dynamics bug

What the true-anomaly → mean-anomaly conversion, called only after a burn, had its $\sqrt{1 \pm e}$ factors swapped. Each burn off an apse silently moved the chaser along-track (up to 700 km at $e = 0.05$); a median 25° of free phasing per episode. Burns at apsides were unaffected, so every unit test passed.

Found twice on one day: by formula inspection while porting the dynamics into the navigation code, and by the fuel audit, where 15 % of episodes beat half the two-impulse optimum on the same clock.

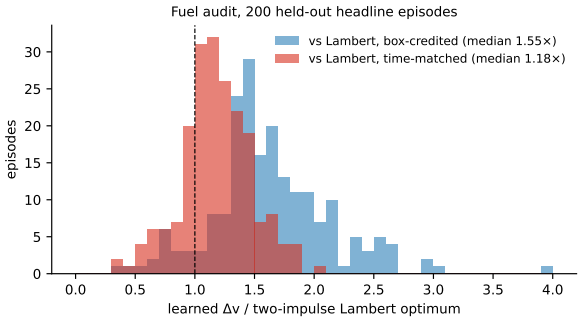
Cost the then-canonical checkpoint went from 97.5 % to 13 % under corrected dynamics. All pre-fix checkpoints and numbers were retired.

Recovery (each piece ablated)

- potential shaping on the *mean-longitude* gap with $\gamma_{shape} = 1$
- physical-gap initialisation and a fixed phase observation
- an episode-clock observation; cap reward 0, not -10
- scripted GNC expert as a reference: 99.2 %

Lesson the comparator caught what the success metric could not. Audits against physics belong in the training loop.

A5 Fuel audit against a two-impulse optimum

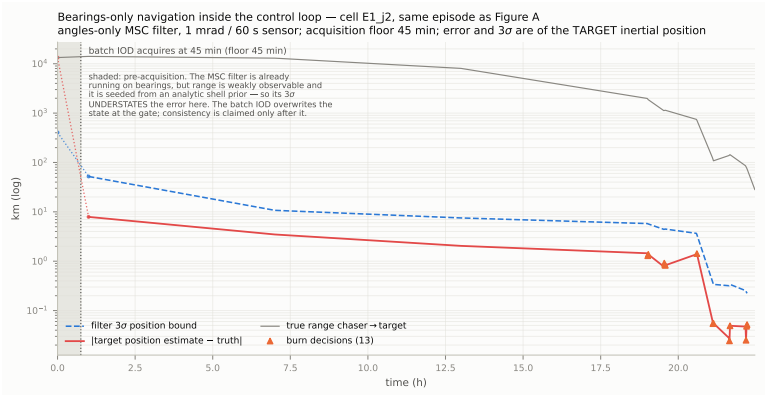


For each episode an offline solver finds the cheapest two-burn transfer from the chaser's start to the target's position at arrival, searching over how long to wait before the first burn and how long to fly, on the same clock the policy used.

Median learned Δv is **1.18**$\times$ that optimum (time-matched); 1.55$\times$ when the optimum is credited the box's 50 m/s velocity tolerance.

200 held-out episodes. Ratios below 1 happen because arrival is credited inside a box, not at an exact state, and a two-burn plan is not the unconstrained optimum.

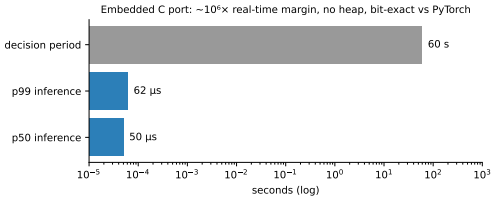
A6 Navigation in one picture: blind, acquire, converge

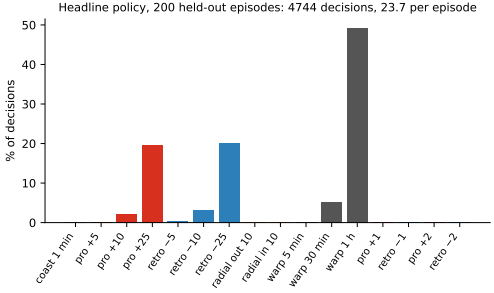


The chaser measures only the direction to the target, not the distance. One direction tells you nothing about range; a sequence of directions while both orbits curve does. A batch solver tries many ranges along the first line of sight, propagates each, and keeps the one whose predicted directions match what was seen. That hands a full state to a recursive filter that keeps refining it.

Policy result: trained on truth and handed the estimate, **51 %**; trained with the filter in the loop, **97 %** (2-D tight box) and **46→98 %** across three seeds in 3-D. It learned to wait for acquisition and burn nothing while blind (median Δv before acquisition 75→0 m/s).

A7 C port of the policy





- The trained weights exported to a C header; forward pass written by hand: linear, GELU, LSTM cell, two linear heads.
- No heap, no globals; the caller owns the 1 kB of recurrent state; fixed work per call.
- Checked against PyTorch: identical action on 7,048 recorded observations and an identical action stream over 200 closed-loop episodes.
- Median 50 μs per decision on a laptop against a 60 s decision period (a benchmark, not a worst-case timing).
- Bottom: what the headline policy actually does, per decision: warps and prograde/retrograde 25 m/s dominate.

A8 FAQ

Why RL, when Lambert exists? Lambert needs perfect state and gives an open-loop two-burn plan; the policy is closed-loop on an estimate, respects a budget, and one network covers seven regimes. Lambert is the grader, not the competitor.

Why discrete actions? PPO is stable with them; a continuous head collapsed early on. The grid costs fuel (1.14–1.18× across two evaluation sets); finer grades are the obvious next step.

Why not MPC? It is the right production baseline and the comparison I would run next, with the same filter in the loop.

Is 200 episodes enough? At 97 % the 95 % Wilson interval is about ±2.5 pp; for 0/200 failures the lower bound is 98.2 %. I report intervals, not "100 %".

Single seed? The six-skill policy reproduced at a second training seed (tight box 92.5 %, wide cells 93.5–100 %). The seventh skill's four-step refinement was run once by design; its first two steps reproduced at a second seed with matching gains. The 2-D headline (5), wide envelope (4), 3-D nav and J2 rungs (3) are multi-seed.

Why coast 24 h? J2 nodal precession closes a plane gap for free; burning it would exceed the tank. The policy learned to wait.

How does the filter work? Angles only, so range comes from how the bearing changes while the orbits curve; a batch solver picks the range that explains the first hour of bearings, then a recursive filter refines it. The finding I own is on the policy side: train on the estimate or lose 46 points.

What would you do differently? Audit against physics from day one; log learning curves on every run; design the observation with the filter in mind from the start.